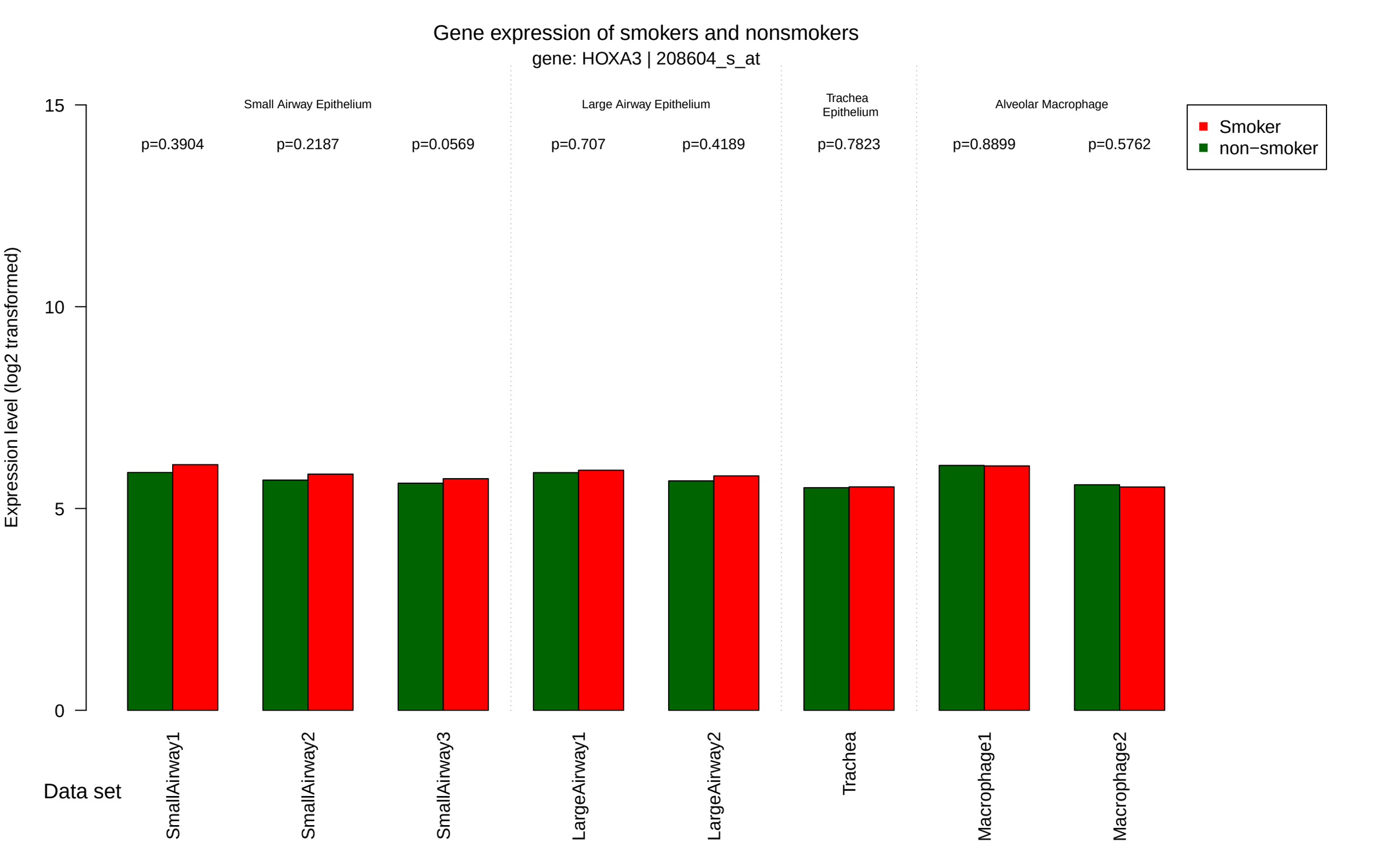
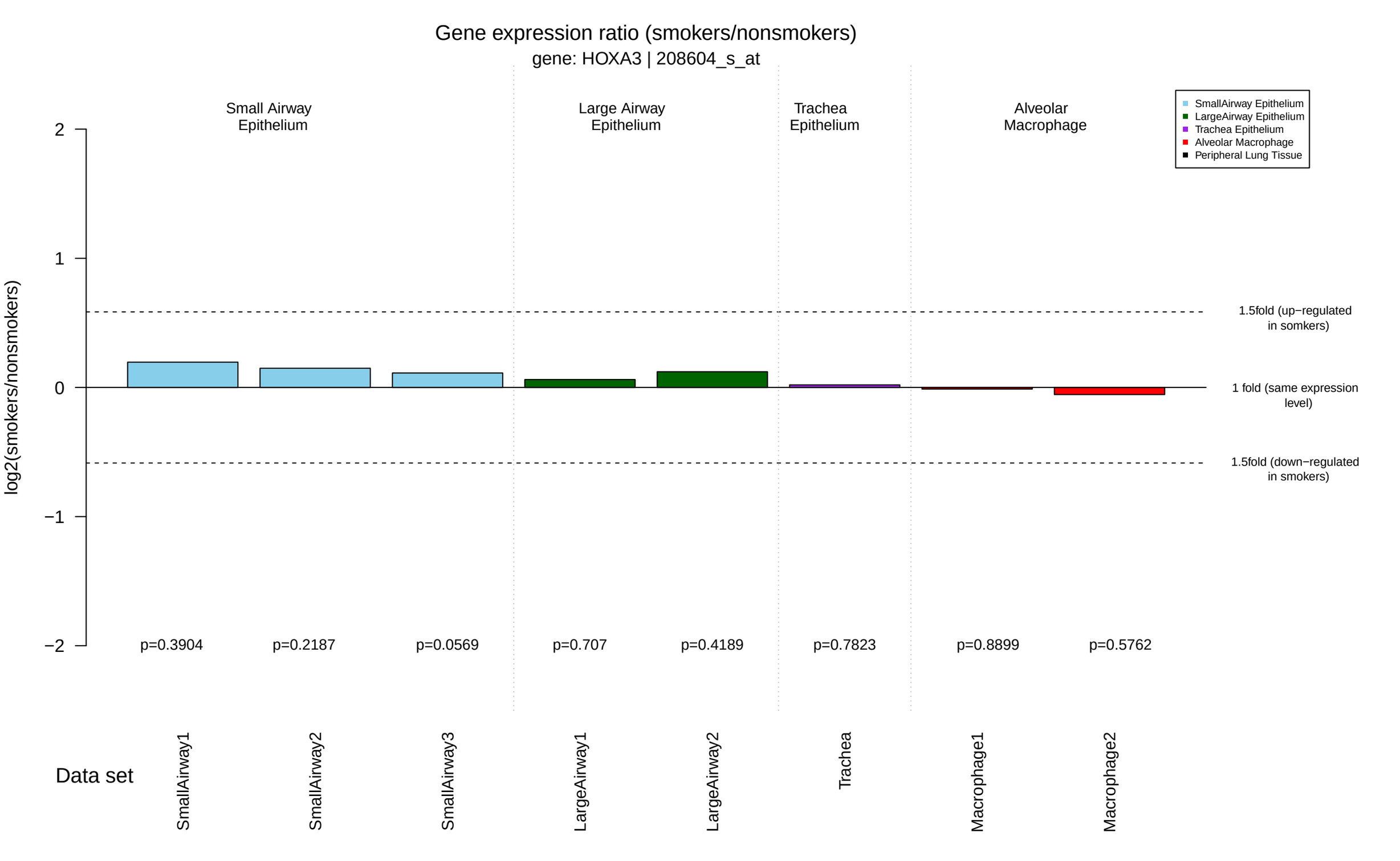
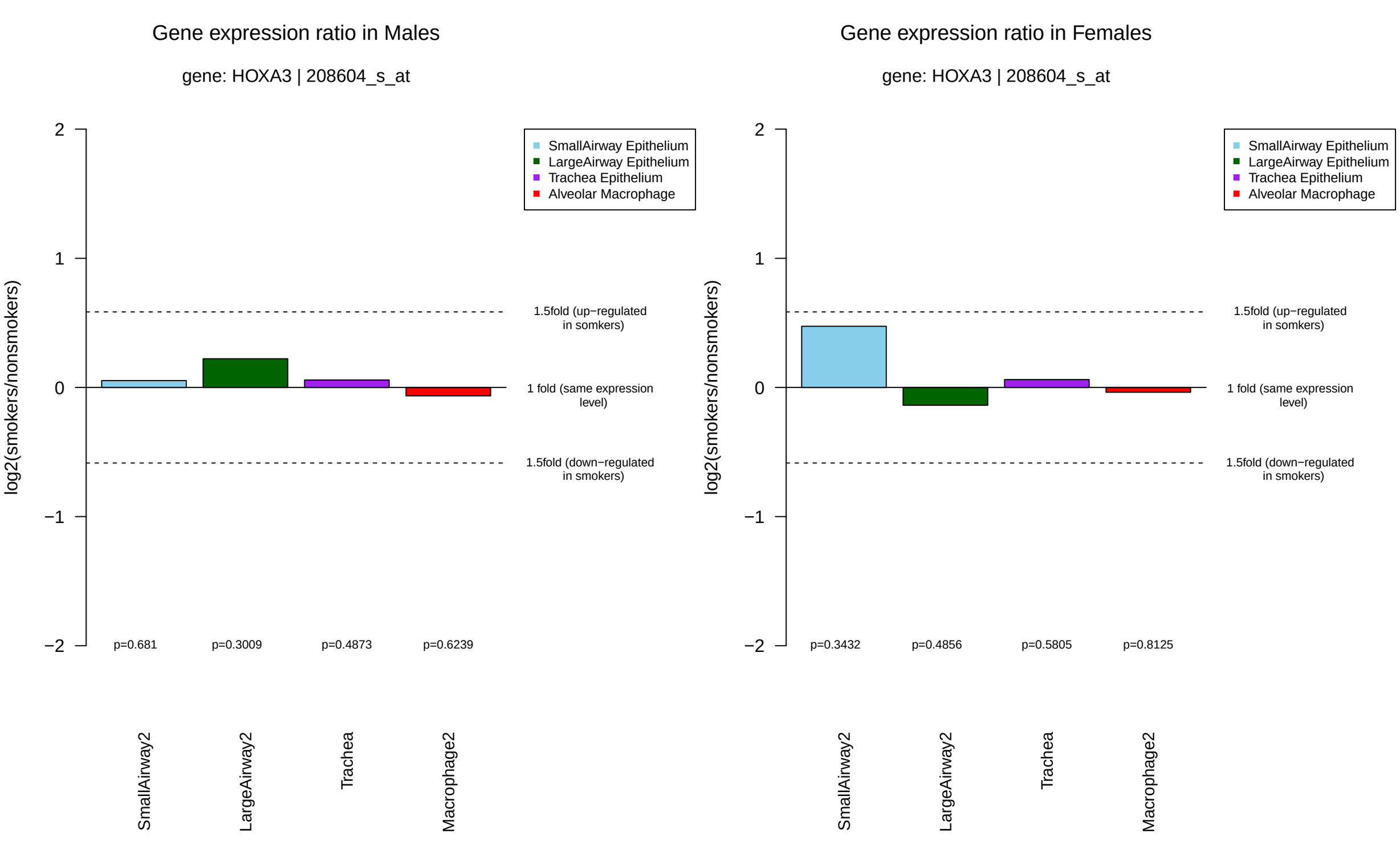
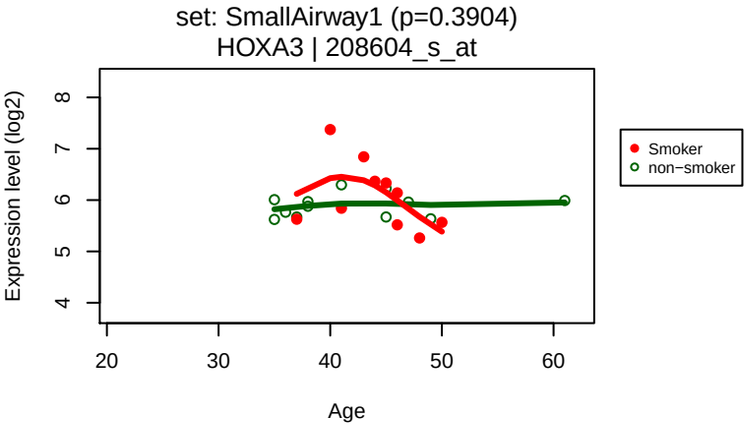
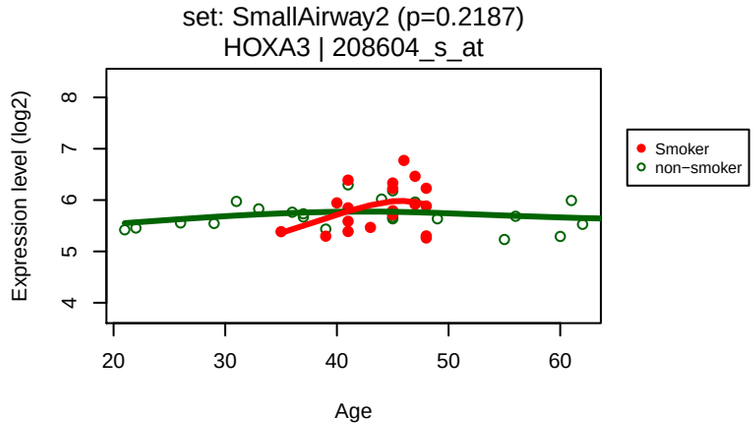
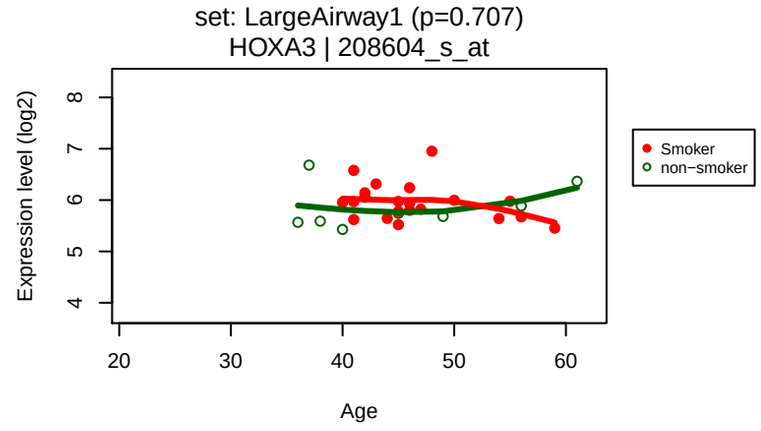

Gene expression of smokers and nonsmokers
gene: HOXA3 | 208604_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3904
p=0.2187
p=0.0569
p=0.707
p=0.4189
p=0.7823
p=0.8899
p=0.5762
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

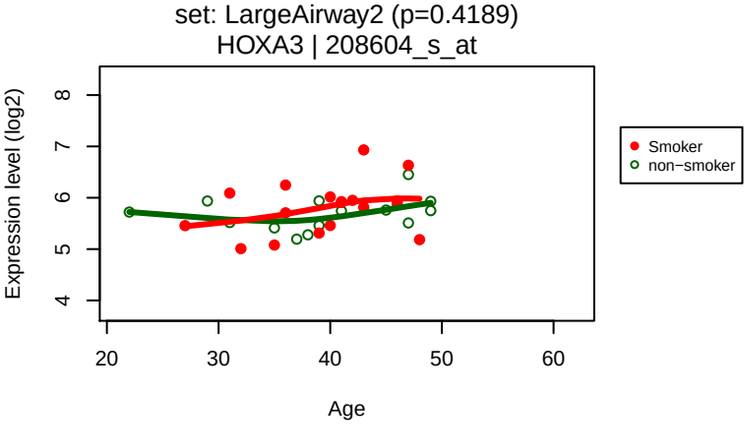

Gene expression ratio (smokers/nonsmokers)
gene: HOXA3 | 208604_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3904
p=0.2187
p=0.0569
p=0.707
p=0.4189
p=0.7823
p=0.8899
p=0.5762
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

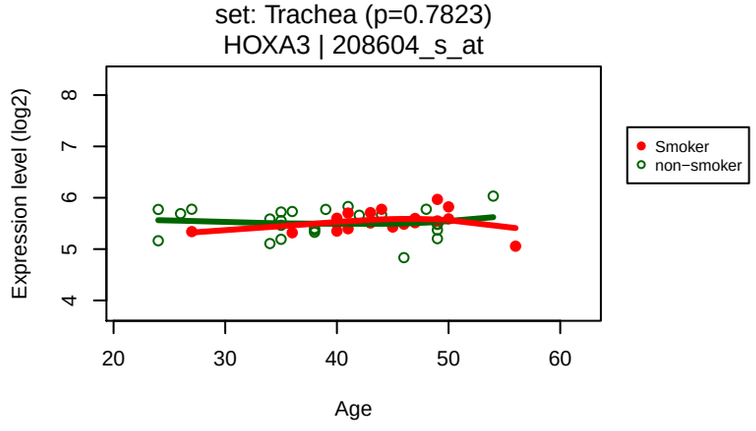

Gene expression ratio in Males
gene: HOXA3 | 208604_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.681
p=0.3009
p=0.4873
p=0.6239
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA3 | 208604_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3432
p=0.4856
p=0.5805
p=0.8125
SmallAirway2
LargeAirway2
Trachea
Macrophage2

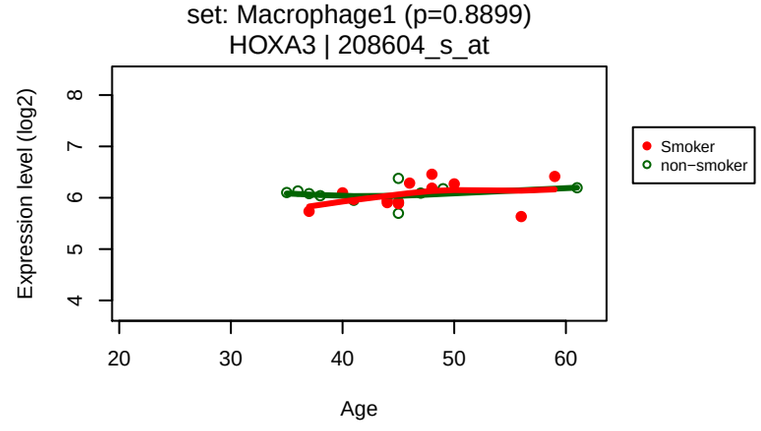

set: SmallAirway1 (p=0.3904)
HOXA3 | 208604_s_at
Expression level (log2)
Age
Smoker
non−smoker

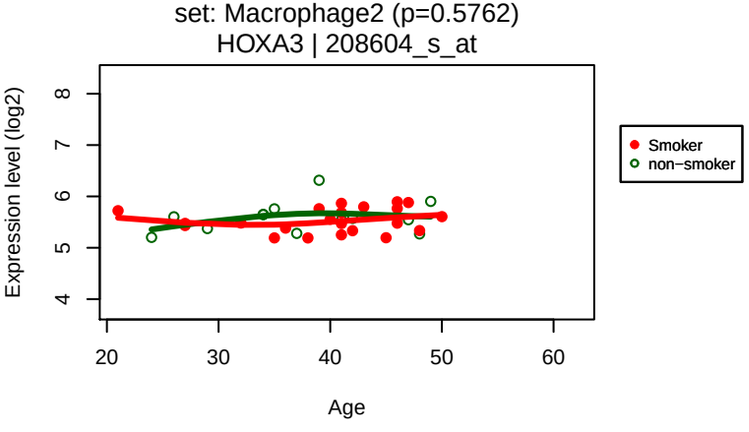

set: SmallAirway2 (p=0.2187)
HOXA3 | 208604_s_at
Expression level (log2)
Age
Smoker
non−smoker

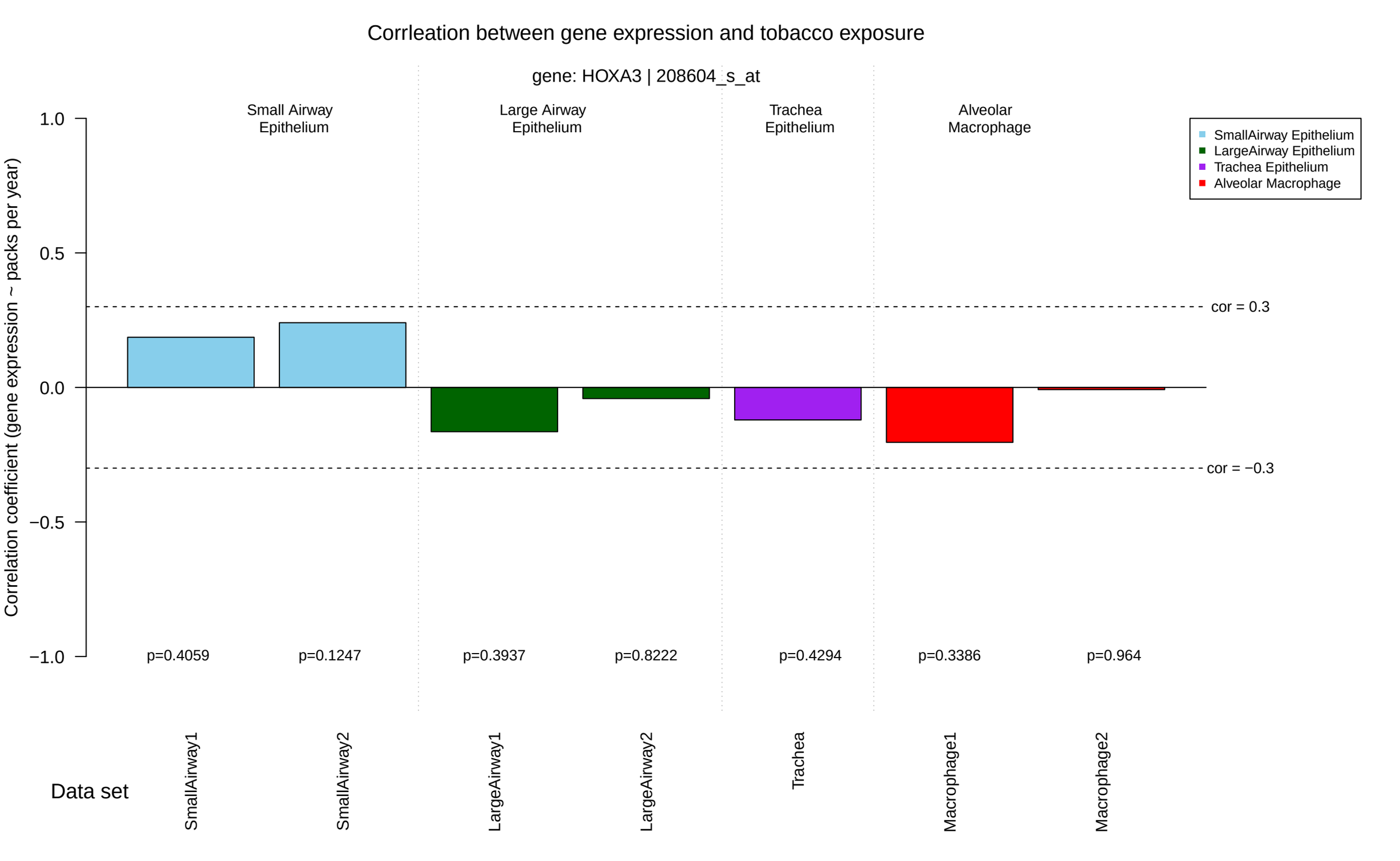

set: LargeAirway1 (p=0.707)
HOXA3 | 208604_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4189)
HOXA3 | 208604_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7823)
HOXA3 | 208604_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8899)
HOXA3 | 208604_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5762)
HOXA3 | 208604_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA3 | 208604_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4059
p=0.1247
p=0.3937
p=0.8222
p=0.4294
p=0.3386
p=0.964
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2